UNIVERSITY OF CAPE COAST
DEPARTMENT OF MATHEMATICS
MAT 101: ALGEBRA AND TRIGONOMETRY
TUTORIAL SET III

1. Maximize the objective function $Z = 3x + y$ subject to the constraints $2x + y \leq 8,\ \ 2x + y \leq 12,\ \ x \geq 0,\ \ y \geq 0.$

2. (a) Maximize $P = 5x + 6y$ subject to
   $x + y \leq 80,\ \ 3x + 2y \leq 220,\ \ 2x + 3y \leq 210,\ \ x, y \geq 0.$
   (b) Minimize $Z = x + y$ subject to
   $x - y \geq 0,\ \ 4x + 3y \geq 12,\ \ 9x + 11y \leq 99,\ \ x \leq 8,\ \ x, y \geq 0.$

3. If a, b, c are positive real numbers,
   (a) show that $\log_b a = \frac{\log_c a}{\log_c a}$; hence or otherwise find :
      i. the values of x for which $2\log_4 x + 9\log_x 4 = 9.$
      ii. the limit as n tends to infinity of $\log_8(12n - 5) - \log_2(3n + 4)^{\frac{1}{3}}.$

4. (a) Given that $\log_b a + \log_c a = 2\log_b a \log_c a$, express a in terms of b and c.
   (b) Solve the simultaneous equation :

$$\log_y x - 4\log_x y = 0$$
$$\log_{10} x + \log 10y = 1$$

5. A sequence is defined by statements; $U_1 = 1, U_{n+1} = U_n + (\frac{1}{4})^n, \forall n \geq 1.$
   (a) Write down U_3 and U_4.
   (b) Find an expression for U_n in terms of n only.
   (c) Evaluate $\lim_{n\to\infty} U_n$.

6. A sequence is defined by the statements; $U_1 = \frac{5}{2}, U_{n+1} = 2 - \frac{1}{3}U_n, \forall n \geq 1.$
   (a) Show that the sequence may also be defined by the relation $U_{n+1} = U_n + 4\left(\frac{-1}{3}\right)^n, \forall n \geq 1.$
   (b) Deduce an explicit expression for U_n
   (c) Find $\lim_{n\to\infty} \frac{S_n}{n}$, where S_n is the sum of the first n terms of the sequence.

7. (a) Find the first four terms in the binomial expansion of each of the following, stating the values of x for which each expansion is valid
   i. $(1+x)^{\frac{1}{3}}$
   ii. $(1+x)^{-2}$
   iii. $(4+x)^{\frac{1}{2}}$
   iv. $\left(2-\frac{1}{2}x\right)^{7}$

   (b) Use a suitable expansion in (a) to calculate to four decimal places
   i. $\sqrt[3]{1.06}$
   ii. $\sqrt{4.05}$

8. (a) Sketch the curve of $f: x \to \frac{2x}{x^2-4}$; showing clearly all the asymptotes and intercepts, if any on the axes

   (b) Decompose the function in (a) into partial fractions.

   (c) Hence, find the expansion of $f(x)$ as far as the term in x^5.

   (d) Show that $f(x)$ may be defined as $f(x) = -\frac{1}{2}\sum_{r=0}^{\infty}\frac{x^{2r+1}}{4r}$.

9. The nth term of the series $5+5+6\frac{1}{3}+\ldots$ is given by $U_n = a(\frac{1}{3})^{n-1}+bn$, where a and b are constants. Find;

   (a) the values of a and b.

   (b) an expression for the sum of the first n-terms of the series, S_n, hence find S_8.

10. The nth term of the series $1+\frac{1}{6}+\frac{1}{72}+\ldots$ is given by $U_n = a\left(\frac{1}{4}\right)^n + b\left(\frac{1}{3}\right)^n$.

    (a) find the values of a and b.

    (b) obtain a formula for the sum of the first n terms of the series, and find its limiting value.